

EU AI Act Ontology Report

Proof-of-concept OWL 2 DL ontology (ontology/aia-ont.ttl, namespace aia:).

1. Scope

Version. The Act as amended by Regulation (EU) 2026/1744 (consolidated OJ text, 27 July 2026).

Coverage:

- the four risk tiers: prohibited, high-risk, transparency-required, minimal
- the two ways a system becomes high-risk (Art. 6): as a product, or safety component of one, under Annex I legislation that requires third-party assessment; or by an Annex III use case
- the ten prohibited practices of Art. 5 and their exemptions
- provider (Art. 16) and deployer (Art. 26) duties; when another party becomes provider (Art. 25)
- who supervises each duty, and how a high-risk system's conformity is assessed (Art. 43): by the provider itself, by a notified body, or under its Annex I sector law

Excluded: general-purpose AI models (Ch. V), Art. 50 transparency duties, Art. 27 impact assessments.

Existing ontologies.

Imported:

- [DPV eu-aiact v2.3](#): risk tiers, Annex I and III lists, authorities
- [AIRO 1.0](#): AI systems and operators
- [ELI 1.4](#): the Act's articles as citable legal resources
- [SKOS](#): The task asks for a label and description on every term, and eu-aiact uses skos:prefLabel and skos:definition.

Aligned: [PROV-O](#) and [ORG](#) model an agent acting in a role, as our role assignments do; [DPV core](#) is what eu-aiact extends; [ODRL](#) is the standard vocabulary of duties and permissions.

Added. Seventeen local classes to meet the competency questions:

- role assignment, reassignment and termination: aia:Role, RoleAssignment, RegulatoryEvent, RoleReassignment, ReassignmentCondition, TerminatedRoleAssignment
- classification grounds: ClassificationGround, PurposeGround, AnnexIIIGround, DerogatedAnnexIIIGround, AnnexIGround, HighRiskByGround
- duties and prohibitions: aia:Obligation, ProhibitedPractice, Exemption
- oversight and conformity: OversightAssignment, ConformityAssessmentRoute

A worked example (aia-example.ttl) covers all four tiers (see README.md for details).

2. AI-assisted pipeline

A Claude Project (**Opus 5.0**) holding the task specification was used to plan the deliverables and draft a survey of existing ontologies, competency questions, SPARQL and SHACL shapes. Claude Code (Opus 5.0 and 5.5) was used to generate the ontology and ran first-pass validation in parallel with my verification in intermediate steps; Claude Code ran HermiT reasoning and SHACL validation in a command shell as part of its development process. I then manually loaded the ontology into Protégé and ran HermiT for validation (see screenshots in evidence/).

Nothing Claude proposed was finalised until checked against **parsed RDF** (claims about an ontology) and **the Official Journal** (the Act).

3. Key modelling decisions

Roles are reified. A role is held per system: under Article 25(1)(b) a deployer that substantially modifies a high-risk system becomes the provider of that system, and Article 25(2) ends the original provider's role for it alone.

aia:RoleAssignment therefore links actor, role and system.

Obligations attach to roles. The Act addresses duties to roles: Article 16 lists what providers must do and Article 26 what deployers must do, so a party's duties follow from the roles it holds. No surveyed ontology links a duty to the party bound by it, so aia:borneBy links each obligation to its role, and aia:borneOnTermination carries the duties Article 25(2) leaves the former provider.

Annex III is split in two. Article 6(3) takes some Annex III systems back out of high risk, and OWL cannot retract an inference, so the exception is recorded as a separate class: aia:AnnexIIIGround makes a system high-risk, aia:DerogatedAnnexIIIGround does not. An AI system acting as a recruitment tool that ranks candidates is high-risk; one that only reformats CVs is a narrow procedural task and is not.

Article 5 prohibits practices, not systems. What is banned is how a system is used, not the system itself: an emotion-recognition system is prohibited in the workplace or education (Art. 5(1)(f)) but allowed there for medical or safety reasons. aia:ProhibitedPractice therefore models the use, separately from eu-aiact:ProhibitedAISystem.

4. Competency questions and SPARQL

1. Into which risk category does a system fall, and on what grounds — intended purpose, Annex III area, or Annex I product route?
2. Which practices are prohibited, which article prohibits each, and which have conditional exemptions?
3. Which obligations attach to the provider of a high-risk system, and which to its deployer?
4. Under what conditions does a deployer, distributor or importer become a provider, which obligations does it acquire, and what becomes of the initial provider?
5. Which authority oversees a given obligation, and which conformity assessment route applies?

The queries for all five are in queries/. **CQ3**, below, shows a provider whose role ended keeping only its residual duties.

```
SELECT ?actorLabel ?roleLabel ?standing ?obligationLabel ?article WHERE {
  VALUES ?system { aia:ExampleRecruitmentScreeningSystem }
  ?system aia:hasRiskCategory eu-aiact:RiskLevelHigh .
  ?a a aia:RoleAssignment ; aia:concernsSystem ?system ;
    aia:hasRole ?role ; aia:heldBy ?actor .
  ?actor skos:prefLabel ?actorLabel . ?role skos:prefLabel ?roleLabel .
  FILTER (?role IN (aia:Provider, aia:Deployer))
  OPTIONAL { ?a a aia:TerminatedRoleAssignment . BIND ("ended" AS ?standing) }
  ?obligation a aia:Obligation ; aia:appliesToCategory eu-aiact:RiskLevelHigh ;
    skos:prefLabel ?obligationLabel ; aia:statedIn ?article .
  FILTER ( (!BOUND(?standing) && EXISTS { ?obligation aia:borneBy ?role })
    || ( BOUND(?standing) && EXISTS { ?obligation aia:borneOnTermination ?role }) )
} ORDER BY ?actorLabel ?roleLabel ?article
```

5. Validation and evaluation metrics

The ontology loads in Protégé and is consistent under Hermit (see screenshots in evidence/). Proposed metrics:

- **CQ answerability:** all five competency questions return answers over the worked example.
- **Annotation completeness:** every term has a label, a definition and an article reference.
- **Provision coverage:** modelled items against the Act's own lists, e.g. all ten Art. 5 practices.
- **Expert legal review** of each obligation and practice, the only check that catches a wrong reading of the law.